

# WASP-43b

R-0597

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_WASP-43\_230115 · created 2026-07-25T01:54:23+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

## Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2459959.8599 +/- 0.0023 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.197 +/- 0.057                 |
| Transit depth (Rp/Rs)^2                           | 3.9 +/- 2.26 [%]                |
| Orbital Inclination (inc)                         | 86.32 +/- 2.93                  |
| Airmass coefficient 1 (a1)                        | 16.46 +/- 10.85                 |
| Airmass coefficient 2 (a2)                        | -2.08 +/- 0.55                  |
| Scatter in the residuals of the lightcurve fit is | 46.8 %                          |
| Best Comparison Star                              | #3 - [504, 384]                 |
| Optimal Aperture                                  | 2.99                            |
| Optimal Annulus                                   | 3.63                            |
| Transit Duration (day)                            | 0.06 +/- 0.0058                 |

## Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R■ fit vs published    | 0.197 ± 0.057 vs 0.1594 (deviation 0.66σ) |
| Duration fit vs published | 0.06 d vs 0.0512 d (deviation 1.51σ)      |
| Mid-time O–C              | -2.9 min                                  |
| Depth SNR                 | 3.5                                       |
| Residual scatter          | 46.8 %                                    |

## Light curve

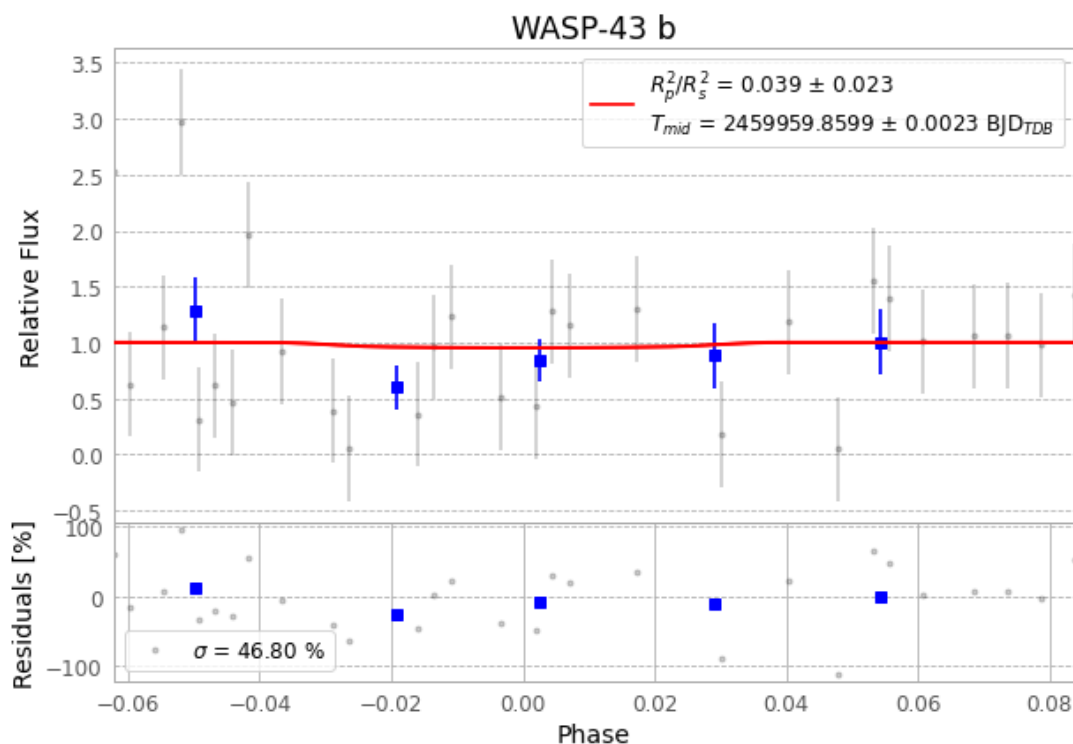


Figure 1 — FinalLightCurve\_WASP-43 b\_2023-01-15.png (SHA-256 13f37ca0aab991dc...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [584, 178], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 4ec6c7287f3a3698...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit b365494

## Data provenance

63 FITS frames (41 MB), WASP-43230115070315.FITS → WASP-43230115100915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-230115.log (93cc56e17cca...), FinalLightCurve\_WASP-43 b\_2023-01-15.png (13f37ca0aab9...), FinalLightCurve\_WASP-43 b\_2023-01-15.csv (226f365c079d...)

## Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 01:54Z.